Submission Summary

Conference Name

1st Multidisciplinary International Conference on Advancing Agentic AI Paradigm through Neuro-Symbolic AI, Machine Learning, Deep Learning, and Swarm Intelligence

Paper ID

120

Paper Title

Text Mining and Sentiment Analysis Based Intelligent System for Monitoring and Evaluating Public Health Awareness Campaigns

Abstract

The COVID-19 pandemic generated an unprecedented volume of social media discourse, creating both an opportunity and a challenge for public health authorities seeking to monitor public sentiment and counter misinformation in real time. This paper presents a unified text mining and sentiment analysis framework designed for public health awareness campaign monitoring. The system is evaluated on a large-scale dataset of 409,915 COVID-19 tweets collected across three temporal windows spanning April 2020 to June 2021. A Long Short-Term Memory (LSTM) deep learning model is trained for three-class sentiment classification — negative, neutral, and positive — achieving a test accuracy of 98.79% and a macro-average F1-score of 0.99, outperforming comparable studies in the literature. A complementary TF-IDF and Logistic Regression pipeline for misinformation detection attains 99.95% accuracy using a domain-adapted keyword lexicon. Beyond classification, the system incorporates statistical temporal spike detection based on a two-standard-deviation threshold, engagement-weighted sentiment scoring, geographic regional analysis, and a rule-based policy recommendation engine grounded in health communication principles. The entire pipeline is encapsulated within an interactive Streamlit web dashboard enabling real-time single-tweet analysis and bulk campaign monitoring. Eight negative sentiment spike dates were detected and found to align with documented pandemic events. The system correctly recommended a targeted fact-check strategy based on the corpus-level

negative sentiment rate of 26.74%. This work addresses a critical gap between NLP research and deployable health communication tools.

Created
2026-04-22 17:36:52 GMT+5:30

Last Modified
2026-04-22 17:36:52 GMT+5:30

Authors
Mokshagna Ratakonda (SRM Institute of Science and Technology)
<ratakondamokshagna@gmail.com>
Dyana Vardhan Reddy Tiyyagura (SRM Institute of Science and Technology)
<tdvr1234@gmail.com>

Submission Files
final research paper.pdf (192.7 Kb, 2026-04-22 17:36:49 GMT+5:30)